TATA v2

Monte Carlo Investment Valuation Model

TATA Arbitration Claims Portfolio

N = 5,000 paths | Seed = 42 | 2 claims

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Portfolio Executive Summary

Total SOC: ₹143.20 Crore

Total E[Q]: ₹143.20 Crore

Claims: 2 | MC Paths: 5,000

Arb Win Probability: 70%

Quantum E[Q|WIN]: 72.00% of SOC

Reference Scenario: 10% Upfront / 20% Tata Tail (SOC)

E[MOIC]: 3.45×

E[XIRR]: 19.1%

P(Loss): 39.0%

E[Net Return]: ₹66.24 Cr

VaR 1%: ₹-28.72 Cr

P(IRR > 30%): 61.0%

Verdict: Marginal

Claim Portfolio Overview

Claim	Archetype	SOC (₹ Cr)	Jurisdiction	Win Rate	E[Dur] (mo)	E[Q] (₹ Cr)
ORIGO-113-2023	Breach_Of_Contract	₹87.00	DOMESTIC	61.0%	36.3	₹87.00
ORIGO-183-2022	Breach_Of_Contract	₹56.20	DOMESTIC	61.0%	36.3	₹56.20

Probability Tree — Combined Outcomes

Claim	P(TRUE_WIN)	P(RESTART)	P(LOSE)	Final Win Rate
ORIGO-113-2023	61.0%	0.0%	39.0%	61.0%
ORIGO-183-2022	61.0%	0.0%	39.0%	61.0%

Quantum Model

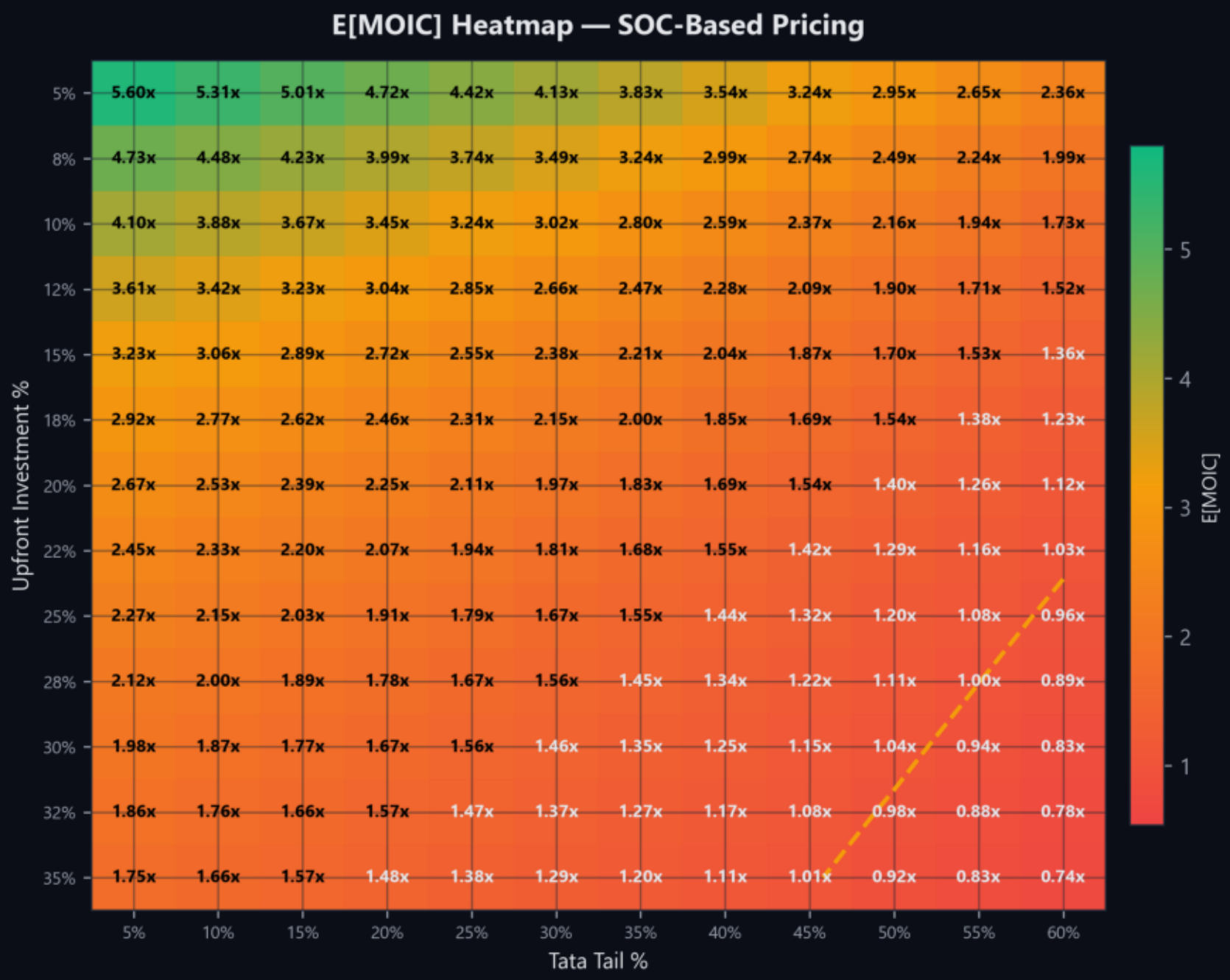
Band	Range (% SOC)	Probability	E[Q] contribution
Band 1	0% – 20%	15%	0.0150
Band 2	20% – 40%	5%	0.0150
Band 3	40% – 60%	5%	0.0250
Band 4	60% – 80%	5%	0.0350
Band 5	80% – 100%	70%	0.6300
TOTAL E[Q|WIN]			0.7200 (72.00% of SOC)
Claim	SOC (₹ Cr)	E[Q] (₹ Cr)	E[Q]/SOC
ORIGO-113-2023	₹87.00	₹87.00	100.00%
ORIGO-183-2022	₹56.20	₹56.20	100.00%

Timeline Analysis — Duration Distributions

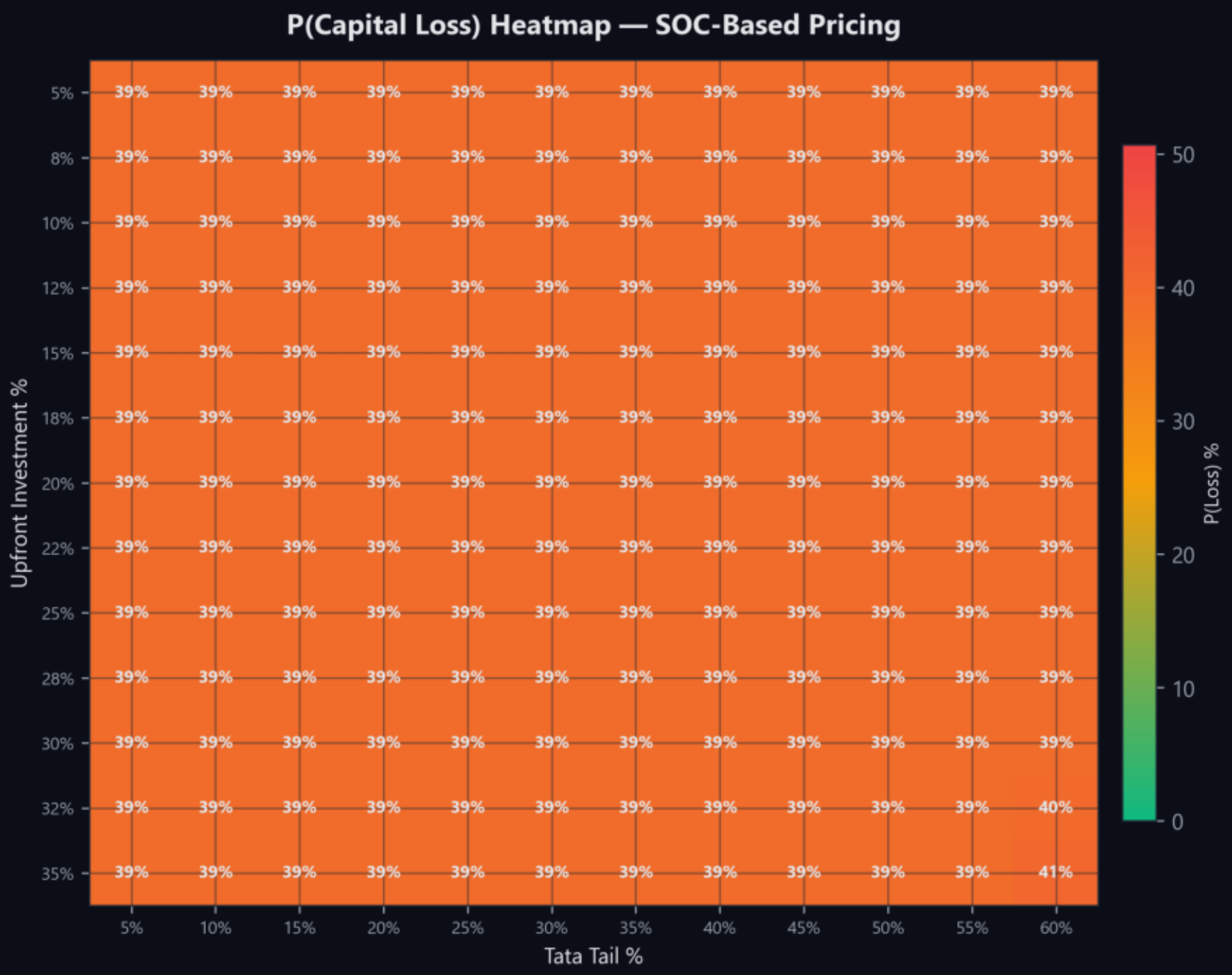
Legal Cost Analysis

Claim	E[Cost] (₹ Cr)	Median	P5	P95
ORIGO-113-2023	₹5.96	₹5.75	₹4.62	₹7.69
ORIGO-183-2022	₹5.96	₹5.75	₹4.62	₹7.69
PORTFOLIO TOTAL	₹11.91			

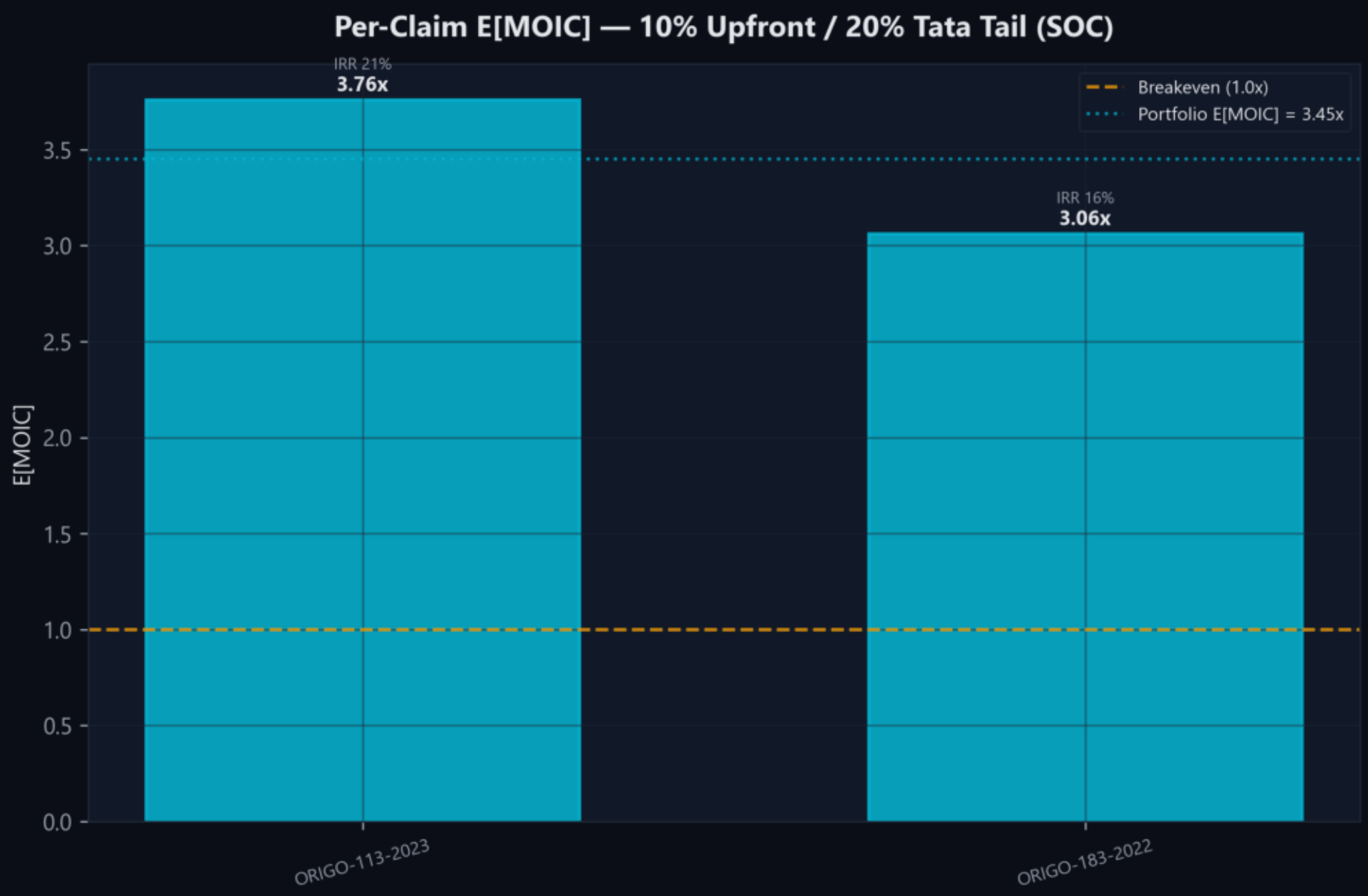
E[MOIC] Heatmap — SOC Pricing



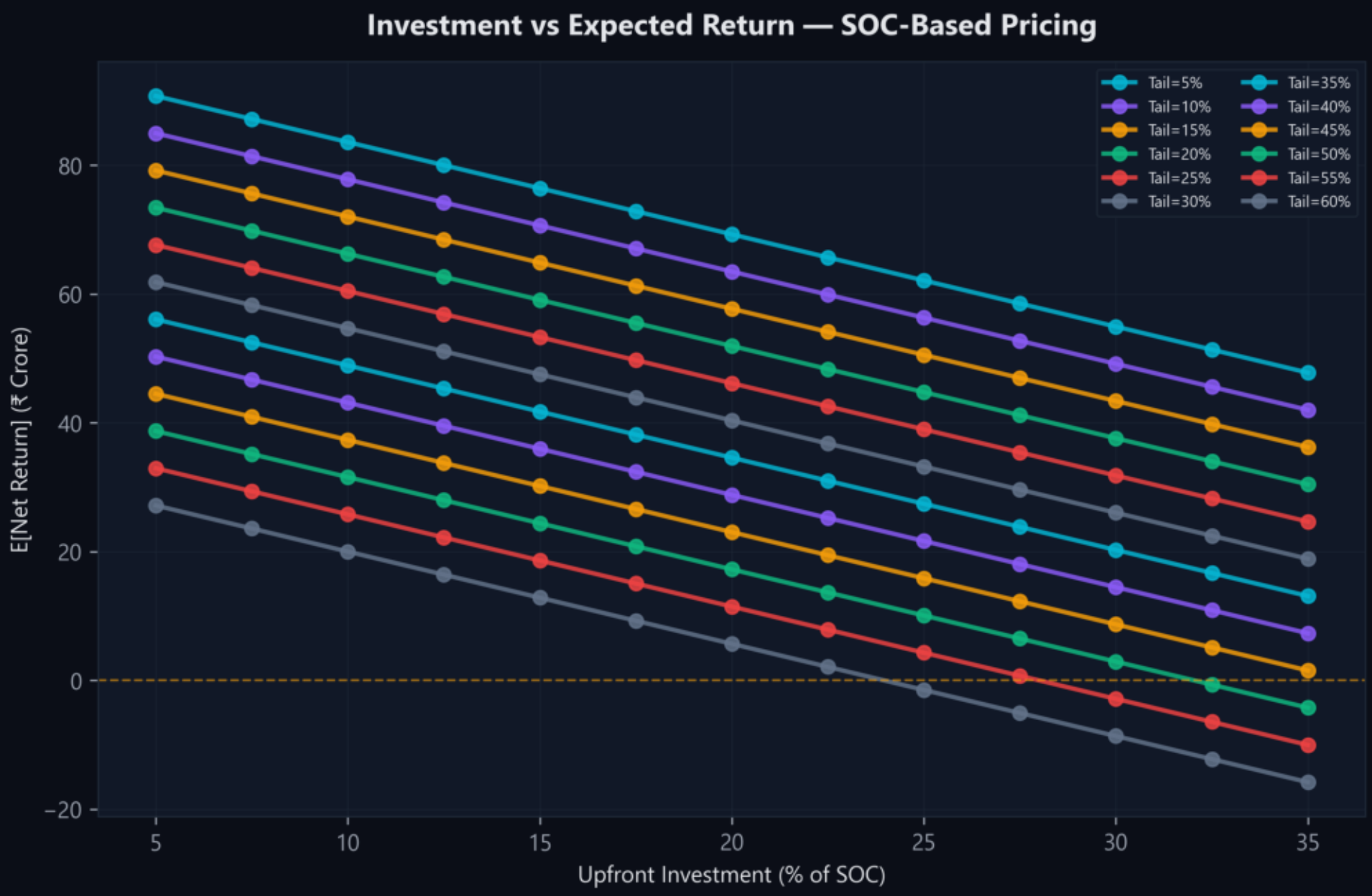
P(Capital Loss) Heatmap



Per-Claim E[MOIC] Analysis



Investment vs Expected Return



Investment Decision Matrix

Investment Decision Matrix — SOC-Based Pricing

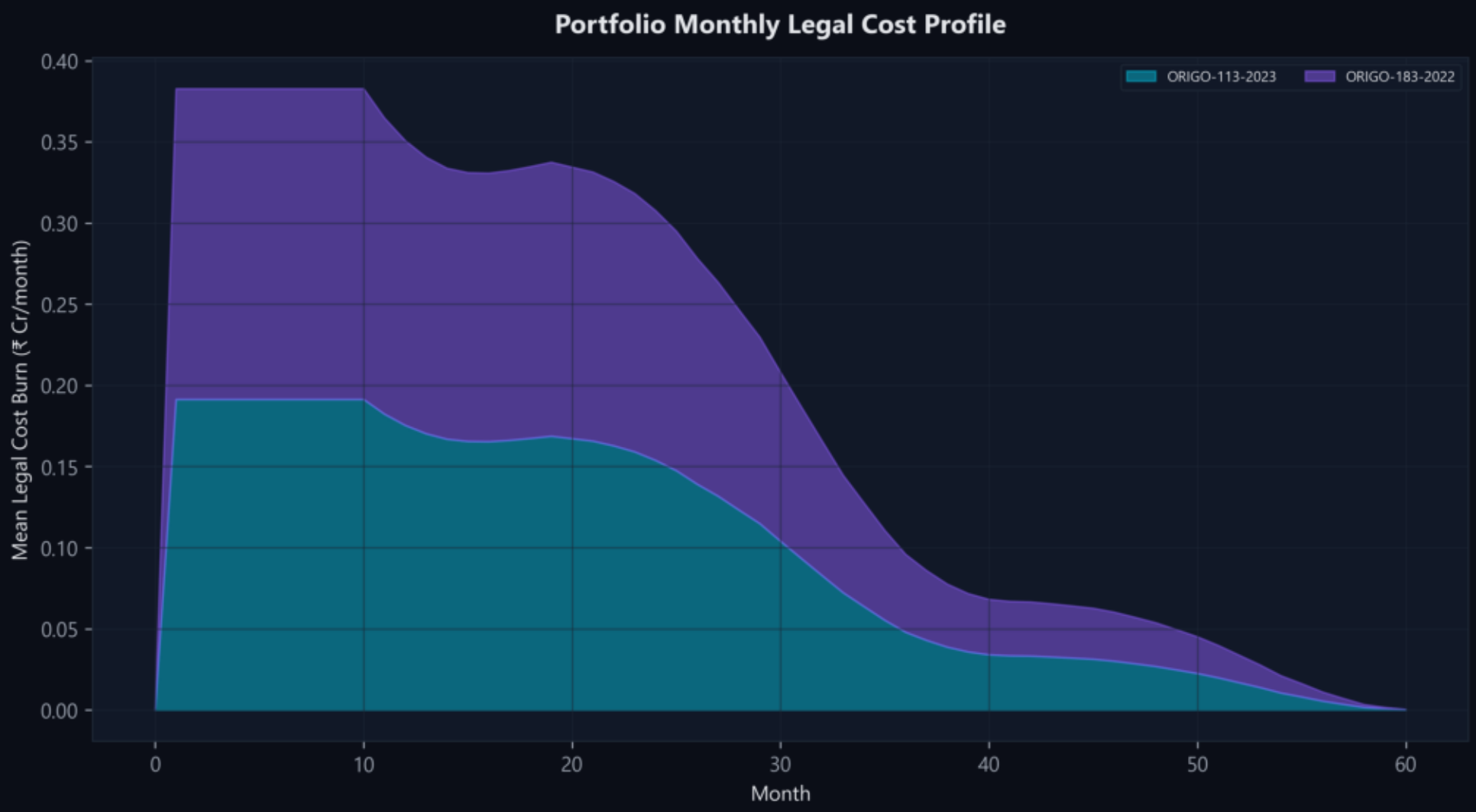
Strong Buy (MOIC>2.5, P(L)<10%)
Attractive (MOIC>1.5, P(L)<25%)
Marginal (MOIC>1.0, P(L)<40%)
Avoid

Upfront Investment % \ Tata Tail %	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%
5%	IRR 52% 5.60x Marginal	IRR 49% 5.31x Marginal	IRR 45% 5.01x Marginal	IRR 42% 4.72x Marginal	IRR 39% 4.42x Marginal	IRR 35% 4.13x Marginal	IRR 31% 3.83x Marginal	IRR 27% 3.54x Marginal	IRR 23% 3.24x Marginal	IRR 19 2.95 Margi	[illegible]	[illegible]
8%	IRR 37% 4.73x Marginal	IRR 34% 4.48x Marginal	IRR 31% 4.23x Marginal	IRR 29% 3.99x Marginal	IRR 25% 3.74x Marginal	IRR 22% 3.49x Marginal	IRR 19% 3.24x Marginal	IRR 15% 2.99x Marginal	IRR 12% 2.74x Marginal	IRR 8 2.49 Marginal	[illegible] Marginal	[illegible] Marginal
10%	IRR 27% 4.10x Marginal	IRR 24% 3.88x Marginal	IRR 22% 3.67x Marginal	IRR 19% 3.45x Marginal	IRR 16% 3.24x Marginal	IRR 13% 3.02x Marginal	IRR 10% 2.80x Marginal	IRR 7% 2.59x Marginal	IRR 4% 2.37x Marginal	IRR 0% 2.16x Marginal	IRR -4% 1.94x Marginal	IRR -8% 1.73x Marginal
12%	IRR 19% 3.61x Marginal	IRR 17% 3.42x Marginal	IRR 15% 3.23x Marginal	IRR 12% 3.04x Marginal	IRR 9% 2.85x Marginal	IRR 7% 2.66x Marginal	IRR 4% 2.47x Marginal	IRR 1% 2.28x Marginal	IRR -2% 2.09x Marginal	IRR -6% 1.90x Marginal	IRR -9% 1.71x Marginal	IRR -13% 1.52x Marginal
15%	IRR 13% 3.23x Marginal	IRR 11% 3.06x Marginal	IRR 9% 2.89x Marginal	IRR 6% 2.72x Marginal	IRR 4% 2.55x Marginal	IRR 1% 2.38x Marginal	IRR -1% 2.21x Marginal	IRR -4% 2.04x Marginal	IRR -7% 1.87x Marginal	IRR -10% 1.70x Marginal	IRR -14% 1.53x Marginal	IRR -17% 1.36x Marginal
18%	IRR 8% 2.92x Marginal	IRR 6% 2.77x Marginal	IRR 4% 2.62x Marginal	IRR 2% 2.46x Marginal	IRR -0% 2.31x Marginal	IRR -3% 2.15x Marginal	IRR -5% 2.00x Marginal	IRR -8% 1.85x Marginal	IRR -11% 1.69x Marginal	IRR -14% 1.54x Marginal	IRR -17% 1.38x Marginal	IRR -21% 1.23x Marginal
20%	IRR 4% 2.67x Marginal	IRR 2% 2.53x Marginal	IRR 0% 2.39x Marginal	IRR -2% 2.25x Marginal	IRR -4% 2.11x Marginal	IRR -7% 1.97x Marginal	IRR -9% 1.83x Marginal	IRR -12% 1.69x Marginal	IRR -14% 1.54x Marginal	IRR -17% 1.40x Marginal	IRR -20% 1.26x Marginal	IRR -24% 1.12x Marginal
22%	IRR 1% 2.45x Marginal	IRR -1% 2.33x Marginal	IRR -3% 2.20x Marginal	IRR -5% 2.07x Marginal	IRR -7% 1.94x Marginal	IRR -10% 1.81x Marginal	IRR -12% 1.68x Marginal	IRR -15% 1.55x Marginal	IRR -17% 1.42x Marginal	IRR -20% 1.29x Marginal	IRR -23% 1.16x Marginal	IRR -26% 1.03x Marginal
25%	IRR -2% 2.27x Marginal	IRR -4% 2.15x Marginal	IRR -6% 2.03x Marginal	IRR -8% 1.91x Marginal	IRR -10% 1.79x Marginal	IRR -12% 1.67x Marginal	IRR -15% 1.55x Marginal	IRR -17% 1.44x Marginal	IRR -20% 1.32x Marginal	IRR -22% 1.20x Marginal	IRR -25% 1.08x Marginal	IRR -28% 0.96x Avoid
28%	IRR -5% 2.12x Marginal	IRR -7% 2.00x Marginal	IRR -9% 1.89x Marginal	IRR -11% 1.78x Marginal	IRR -13% 1.67x Marginal	IRR -15% 1.56x Marginal	IRR -17% 1.45x Marginal	IRR -19% 1.34x Marginal	IRR -22% 1.22x Marginal	IRR -24% 1.11x Marginal	IRR -27% 1.00x Marginal	IRR -30% 0.89x Avoid
30%	IRR -8% 1.98x Marginal	IRR -9% 1.87x Marginal	IRR -11% 1.77x Marginal	IRR -13% 1.67x Marginal	IRR -15% 1.56x Marginal	IRR -17% 1.46x Marginal	IRR -19% 1.35x Marginal	IRR -21% 1.25x Marginal	IRR -24% 1.15x Marginal	IRR -26% 1.04x Marginal	IRR -29% 0.94x Avoid	IRR -32% 0.83x Avoid
32%	IRR -10% 1.86x Marginal	IRR -12% 1.76x Marginal	IRR -13% 1.66x Marginal	IRR -15% 1.57x Marginal	IRR -17% 1.47x Marginal	IRR -19% 1.37x Marginal	IRR -21% 1.27x Marginal	IRR -23% 1.17x Marginal	IRR -26% 1.08x Marginal	IRR -28% 0.98x Avoid	IRR -31% 0.88x Avoid	IRR -34% 0.78x Avoid
35%	IRR -12% 1.75x Marginal	IRR -13% 1.66x Marginal	IRR -15% 1.57x Marginal	IRR -17% 1.48x Marginal	IRR -19% 1.38x Marginal	IRR -21% 1.29x Marginal	IRR -23% 1.20x Marginal	IRR -25% 1.11x Marginal	IRR -27% 1.01x Marginal	IRR -30% 0.92x Avoid	IRR -32% 0.83x Avoid	IRR -35% 0.74x Avoid

Breakeven Analysis — SOC Pricing

Tata Tail %	Max Upfront %	E[MOIC] at Breakeven	E[XIRR] at Breakeven	Verdict
60%	22%	1.03×	-26.2%	Marginal
55%	28%	1.00×	-27.2%	Marginal
50%	30%	1.04×	-26.3%	Marginal
45%	35%	1.01×	-27.2%	Marginal
40%	35%	1.11×	-24.9%	Marginal
35%	35%	1.20×	-22.8%	Marginal
30%	35%	1.29×	-20.8%	Marginal
25%	35%	1.38×	-18.8%	Marginal
20%	35%	1.48×	-17.0%	Marginal
15%	35%	1.57×	-15.2%	Marginal
10%	35%	1.66×	-13.5%	Marginal
5%	35%	1.75×	-11.8%	Marginal

Portfolio Monthly Legal Cost Profile



Appendix — Methodology & Caveats

Monte Carlo Engine

- N independent paths per claim, each traversing the full pipeline.
- Pipeline: DAB → Arbitration → Court Challenge Tree → Re-Arbitration → Payment.
- Quantum drawn from 5 discrete bands conditional on arbitration WIN.
- Legal costs: per-stage annual burn × duration + stochastic overrun.
- XIRR computed via scipy.optimize.brentq with monthly cashflows.

Probability Trees

- Domestic: 4-level tree (S.34 → S.37 → SLP gate → SLP merits) = 24 paths.
- SIAC: 2-level tree (HC → COA) = 8 paths.
- Sc.A (won arb): TRUE_WIN or LOSE. Sc.B (lost arb): RESTART or LOSE.
- RESTART outcomes trigger independent re-arbitration (P(win)=70%).

Investment Model

- Upfront investment = upfront_pct × SOC (or E[Q] for EQ pricing).
- Legal costs funded throughout case timeline.
- Tata tail = TATA's retained % of received quantum.
- MOIC = total_return / total_invested; P(loss) = P(MOIC < 1.0).

Important Caveats

[!] TPL shares set to 100% — actual JV shares are UNCONFIRMED.

[!] Quantum bands & arb win prob are expert-judgment calibrated.

[!] Collection efficiency = 100% (no sovereign default modelled).

[!] IRR undefined for zero-return paths (set to -100%).